HORIZONTAL TAILPLANE PROJECT REPORT

Parametric APDL-Based Structural Modelling and Verification of a Horizontal Tailplane under EASA CS-25 Flight Loads

submitted to

15016606 AIRCRAFT STRUCTURE
ASSOC. PROF. DR. SUWIN SLEESONGSOM

By

ARCHAWIN	**CHINDARATTANAPORN**	**ID NO. 67130043**
ARYAN	**SAJITH**	**ID NO. 67130044**
THANATCHAI	**KHUNPILUKE**	**ID NO. 67130036**
SIRASAK	**PERMSUP**	**ID NO. 67130029**
PEERADOL	**SINGSAMRAN**	**ID NO. 67130061**

Aeronautical Engineering Program
International Academy of Aviation Industry
King Mongkut's Institute of Technology Ladkrabang

October 20, 2026

TABLE OF CONTENT

1. INTRODUCTION 1
- 1.1 Background 1
- 1.2 Problem Statement 1
- 1.3 Scope of the Study 1

2. LABORATORY AIM 2
- 2.1 Objectives 2
- 2.2 Success Criteria 2

3. RELATED THEORIES 3
- 3.1 Flight Envelope and Limit Load Factors 3
- 3.2 Longitudinal Trim and Balancing Tail Load 3
- 3.3 Spanwise Load Distribution and Internal Loads 3
- 3.4 NACA Four-Digit Airfoil Geometry 4
- 3.5 Finite Element Modelling of Semi-Monocoque Structures 4
- 3.6 Modal Analysis and Rigid-Body Modes 4
- 3.7 Equilibrium and Mass Consistency Checks 4
- 3.8 Limit Load, Ultimate Load and Design Allowables 5

4. LABORATORY METHODS 6
- 4.1 Laboratory Workflow 6
- 4.2 Regulatory Basis and Reference Aircraft 6
- 4.3 Aircraft and Load Parameters 7
- 4.4 Horizontal Tailplane Geometry 7
- 4.5 Materials 8
- 4.6 Coordinate System, Units and Software 8
- 4.7 Structural Configurations 8
- 4.8 Verification Acceptance Criteria 9
- 4.9 Refinements to Preliminary Assumptions 9

5. PROCEDURES 11
- 5.1 Step 1: Data Collection and Parameter Definition 11
- 5.2 Step 2: Parametric APDL Script Development 11
- 5.3 Step 3: SFBM Load Pre-processing 14
- 5.4 Step 4: Configuration Setup 18
- 5.5 Step 5: Model Verification 19

6. CONCLUSION AND DISCUSSION 22
- 6.1 Conclusion 22
- 6.2 Discussion 22
- 6.3 Limitations 23
- 6.4 Recommendations for Future Work 23

REFERENCES 24

INTRODUCTION

1.1 Background

The horizontal tailplane (HTP) provides longitudinal stability and generates the balancing tail load needed to trim the aircraft about its centre of gravity. It sits at the far end of the fuselage, so its structure must carry large trim loads while remaining as light as possible, which makes it a demanding and weight-sensitive component of a transport airframe.

In a conventional design process, each change of internal layout requires the CAD geometry to be rebuilt, repaired and remeshed before a new finite element analysis can be run. Palumbo et al. [1] showed that this bottleneck can be removed by generating the complete tailplane structure, from geometry to mesh and loads, inside a single ANSYS Parametric Design Language (APDL) script. Their work, based on a civil aircraft tailplane from the HERFUSE project, provides the geometric baseline and the verification method adopted in this report.

1.2 Problem Statement

Two difficulties had to be resolved before the reference workflow could be applied. First, the aerodynamic loads used by Palumbo et al. are subject to industrial confidentiality and are not published. Second, the project proposal initially considered the CS-VLA standard, whose maximum take-off mass of 750–850 kg is far too small for a tailplane with a span of almost 8 m. A consistent load basis therefore had to be reconstructed from public data.

A finite element model is only useful as a design tool if it can be trusted. The question addressed in this report is therefore whether a single parametric script can generate tailplane models with different internal layouts, loaded with flight loads reconstructed from public data, that are connected correctly, carry the right mass and satisfy equilibrium, so that they can serve as a reliable basis for later stress and design studies.

1.3 Scope of the Study

1) One half-span of the tailplane is modelled as a cantilever clamped at the root rib ($y = 0$).
2) Three internal layouts are generated from the same script: 14 ribs, 3 spars and 8 stringers (Configuration A, reference layout), 9-2-8 (Configuration B) and 7-2-6 (Configuration C).
3) Loads are reconstructed for the governing limit load case D_{fwd} (dive speed, forward centre of gravity limit); other envelope cases are identified but not applied.
4) Analyses are free-free modal analysis, static 1g gravity checks and a static solution under the flight load used to audit equilibrium. Stress assessment, buckling and sizing are outside the scope.
5) The composite skin is represented by an equivalent quasi-isotropic shell; the internal structure is aluminium alloy 7075-T6.
6) All models are solved under the ANSYS Student licence, which limits a model to 128,000 nodes.

LABORATORY AIM

2.1 Objectives

The objectives of this project are as follows:

1) To develop a fully parametric APDL model that generates a multi-spar, multi-rib horizontal tailplane without manual GUI intervention.
2) To reconstruct the critical flight loads from public CS-25 and ATR 42-600 data and map them onto the finite element model with verified equilibrium.
3) To verify the finite element models against modal, mass and equilibrium benchmarks from the reference literature.

2.2 Success Criteria

Each objective is judged against a measurable criterion, so that the result of the laboratory can be stated as pass or fail rather than as an impression:

Objective	Success criterion
1 Parametric model	One macro generates all three layouts with a closed, conformal mesh and no manual editing
2 Load reconstruction and mapping	Root reaction equals the applied half-tail load within 0.001%; target root bending moment reproduced
3 Model verification	All three acceptance gates of Section 4.8 passed for every configuration

Table 2.1 Success criteria for the three objectives

RELATED THEORIES

3.1 Flight Envelope and Limit Load Factors

CS-25 [2] defines the combinations of airspeed and load factor that the structure must withstand, plotted as the V-n diagram. For transport aeroplanes the positive limit manoeuvre load factor depends on the design weight W in Newton (N):

$$n_{max} = 2.1 + 24{,}000 / (W + 10{,}000), \qquad 2.5 \le n_{max} \le 3.8 \tag{3.1}$$

and the negative limit is $n_{min} = -1.0$. The left boundary of the envelope is the stall curve, on which the load factor grows with the square of the speed. The 1g stall speed and the manoeuvre speed, at which the stall curve meets n_{max}, are

$$V_{S1} = \sqrt{[2Wg / (\rho_0 S_w C_{L,max})]}, \qquad V_A = V_{S1} \sqrt{n_{max}} \tag{3.2}$$

where ρ_0 is sea-level air density, S_w the wing area and $C_{L,max}$ the maximum lift coefficient. The design cruise speed V_C is the maximum operating speed, and CS-25.335 allows the design dive speed to be taken as $V_D = 1.25\ V_C$. Dynamic pressure $q = \tfrac{1}{2}\rho_0 V^2$ is largest at V_D, which is why dive-speed cases usually govern tail loads. Gust boundaries (CS-25.341) are straight lines from n = 1 at zero speed whose slope depends on the gust velocity and the wing lift-curve slope [8].

3.2 Longitudinal Trim and Balancing Tail Load

In steady flight the wing lift L_w and the tail load L_H together support the aircraft weight multiplied by the load factor, and their moments about the centre of gravity must cancel together with the wing-body pitching moment [8]:

$$L_w + L_H = nW \tag{3.3}$$

$$L_w (x_{cg} - x_{ac})\, \bar{c} + C_{m0}\, q\, S_w\, \bar{c} - L_H\, l_H = 0 \tag{3.4}$$

Here x_{cg} and x_{ac} are the positions of the centre of gravity and the wing aerodynamic centre as fractions of the mean aerodynamic chord $\bar{c}$, C_{m0} is the wing-body zero-lift pitching moment coefficient and l_H is the tail moment arm. Eliminating L_w gives the balancing tail load (positive upward):

$$L_H = [nW (x_{cg} - x_{ac})\, \bar{c} + C_{m0}\, q\, S_w\, \bar{c}] / [l_H + (x_{cg} - x_{ac})\, \bar{c}] \tag{3.5}$$

A forward centre of gravity ($x_{cg} < x_{ac}$) and a nose-down C_{m0} both make L_H negative, so the largest tail download occurs at the forward CG limit and high dynamic pressure.

3.3 Spanwise Load Distribution and Internal Loads

Schrenk's approximation [7] estimates the spanwise lift distribution of a tapered surface as the average of a distribution proportional to the local chord C(y) and an elliptical distribution with the same total lift. For a half-span s carrying a total load P:

$$\text{shape}(y) = \tfrac{1}{2} \left[C(y) / \textstyle\int_0^s C\, d\eta \;+\; (4 / \pi s) \sqrt{(1 - (y/s)^2)} \right], \qquad w(y) = P \cdot \text{shape}(y) \tag{3.6}$$

The internal shear force and bending moment follow by integrating the running load w(y) from the free tip, where both are zero, towards the root:

$$V(y) = P - \textstyle\int_0^y w(\eta)\, d\eta, \qquad M(y) = \int_y^s V(\eta)\, d\eta \tag{3.7}$$

3.4 NACA Four-Digit Airfoil Geometry

The thickness of a symmetrical NACA four-digit section of chord C and thickness ratio t/c is given at the non-dimensional position $\xi = x/C$ by [4]:

$$y_t = 5\,(t/c)\,C\,[0.2969\sqrt{\xi} - 0.1260\xi - 0.3516\xi^2 + 0.2843\xi^3 - 0.1015\xi^4] \tag{3.8}$$

For NACA 0012, t/c = 0.12 and the upper and lower surfaces are mirror images, so the section produces no lift or pitching moment at zero incidence.

3.5 Finite Element Modelling of Semi-Monocoque Structures

A tailplane is a semi-monocoque structure [9]. The skins carry aerodynamic pressure, shear flow and part of the bending stress; spar webs carry the vertical shear; spar caps and stringers carry spanwise bending; and ribs keep the airfoil shape and transfer local pressure to the spars. The thin plates are modelled with shell elements and the slender stiffeners with beam elements [11]:

- **SHELL181** is a four-node shell with six degrees of freedom per node that combines membrane and bending behaviour and includes transverse shear (first-order shear deformation theory).
- **BEAM188** is a Timoshenko beam that accounts for shear deformation and accepts arbitrary cross-sections such as Hat and rectangular profiles; offsets place the section centroid away from the node line.

For linear static analysis the assembled stiffness matrix K relates the nodal displacement vector u to the applied nodal force vector F:

$$K\,u = F \tag{3.9}$$

Shells and beams that share nodes are fully compatible, so no additional constraint equations are needed when the beam lines coincide with shell edges.

3.6 Modal Analysis and Rigid-Body Modes

Free vibration of an undamped structure with mass matrix M is governed by the generalized eigenvalue problem [11], [12]:

$$(K - \omega_r^2\,M)\,\varphi_r = 0, \qquad f_r = \omega_r / 2\pi \tag{3.10}$$

An unconstrained (free-free) body in three dimensions has exactly six rigid-body modes, three translations and three rotations, with zero frequency because they produce no strain energy. Additional zero-frequency modes would reveal a mechanism or a disconnected part, which is why the free-free spectrum is a sensitive check of model connectivity. In a rigid-body mode the displacement field is

$$u(x) = u_0 + \theta_0 \times (x - x_{CG}) \tag{3.11}$$

so any signed displacement component, for example $u_z = u_{z0} + \theta_x(y - y_{CG}) - \theta_y(x - x_{CG})$, is a linear function of position and its contour lines are straight and parallel.

3.7 Equilibrium and Mass Consistency Checks

Two simple physical checks reveal most assembly errors in a finite element model. First, the mass from the element volumes must equal the total of the mass matrix:

$$M = \Sigma_k\,\rho_k\,V_k \tag{3.12}$$

Second, under a uniform acceleration g the root reactions of a clamped model must equal the weight Mg in the direction of the acceleration and vanish in the other two directions. The strength ratio, defined as the allowable stress divided by the largest stress under 1g, should also be greater than one.

3.8 Limit Load, Ultimate Load and Design Allowables

CS-25.301–.305 [2] distinguish limit loads, the largest loads expected in service, from ultimate loads, which are limit loads multiplied by a factor of safety of 1.5. Design allowables are therefore set as

$$\sigma_{allow} = \sigma_{ref} / 1.50 \tag{3.13}$$

where σ_{ref} is a minimum (design) strength rather than a typical handbook value [13], [14]. Although stress assessment is outside the scope of this report, the allowables are defined here because they are part of the material input of the model.

LABORATORY METHODS

4.1 Laboratory Workflow

The work was organized in five steps, each delivering verified inputs to the next (Figure 4.1). The APDL model (Step 2) and the load module (Step 3) were developed in parallel and joined at the configuration setup. The schedule is summarized in Table 4.1.

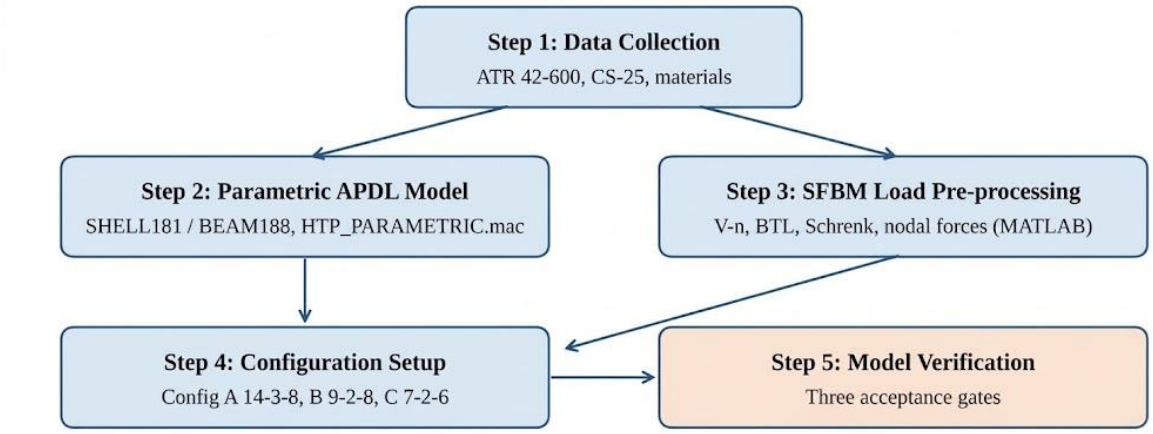


Figure 4.1 Laboratory workflow from data collection to model verification

Task	Description	Step	Dates (2026)	Days
T-01	Initial topic study (aircraft window structure)	Topic selection	16–22 Sep	7
T-02	Change of topic to horizontal tailplane; literature review	Topic selection	23–25 Sep	3
T-03	Data collection and parameter definition	Step 1	26–27 Sep	2
T-04	Parametric APDL script development	Step 2	28 Sep–15 Oct	18
T-05	Proposal presentation for project approval	Milestone	30 Sep	1
T-06	SFBM load pre-processing (MATLAB)	Step 3	5–17 Oct	13
T-07	Configuration setup and model verification	Steps 4–5	17–18 Oct	2
T-08	Final report preparation	Documentation	19–20 Oct	2

Table 4.1 Project task breakdown and schedule

4.2 Regulatory Basis and Reference Aircraft

Applying the CS-VLA mass limits to the HERFUSE tailplane, whose full span is close to 8 m, would create a severe mismatch between the loads and the physical size of the structure. The project therefore moved its regulatory basis to EASA CS-25 [2] and adopted the ATR 42-600 regional turboprop as the reference aircraft. Its publicly documented mass limits, centre of

gravity envelope and design speeds [3] allow the V-n diagram and the balancing tail loads to be synthesized without proprietary data. The tailplane geometry and baseline layout replicate Palumbo et al. [1], while aerodynamic properties missing from the literature, such as the airfoil, are taken from the ATR 42-600.

4.3 Aircraft and Load Parameters

Parameter	Value	Role in the analysis
Aircraft category	CS-25 transport (regional turboprop)	Matches the physical scale of the tailplane
Maximum take-off weight (MTOW)	18,600 kg	Limit load factor and lift at n_{max}
Maximum zero-fuel weight (MZFW)	17,000 kg	Critical wing bending case (no fuel relief)
CG limits	Forward 16.8% MAC; aft 32.5% MAC	Forward limit gives the maximum tail download
Wing area / span	$S_w = 54.5\ m^2$; b = 24.57 m	Aspect ratio and lift-curve slope
Mean aerodynamic chord	$\bar{c} = 2.285$ m	Pitching moment equilibrium
Tail moment arm	$l_H = 11.46$ m	Balancing tail load
Maximum lift coefficient (clean)	$C_{L,max} = 1.50$	Stall speed V_{S1}
Wing-body pitching moment	$C_{m0} = -0.06$	Balancing tail load
Design cruise speed	$V_C = 250$ kt EAS	Envelope boundary; $V_D = 1.25\ V_C$
Factor of safety	1.50	Ultimate load = 1.5 × limit load

Table 4.2 ATR 42-600 mass, aerodynamic and load parameters (CS-25 basis)

4.4 Horizontal Tailplane Geometry

Parameter	Value	Source and application
Airfoil section	NACA 0012 (symmetrical)	ATR 42-600; Equation 3.8
Semi-span	3,978 mm	Literature [1]; length of the FE model
Root chord	3,090 mm	Literature [1]; root attachment dimensions
Tip chord	1,860 mm	Literature [1]; taper ratio 0.602
Dihedral angle	7.0°	Literature [1]; out-of-plane geometry
Maximum thickness-to-chord ratio	12%	Literature [1]; internal space for spars and ribs
Baseline internal layout	14 ribs, 3 spars, 8 stringers	Literature [1]; Configuration A

Table 4.3 Horizontal tailplane geometry

4.5 Materials

Following Palumbo et al., a metallic internal framework is combined with an equivalent isotropic composite skin. Aluminium alloy 7075-T6 is used for spars, ribs, stringers, caps and flanges [5], with the conservative design yield of 480 MPa [13]. The skin is a quasi-isotropic carbon/epoxy laminate [6] with a design strength of 300 MPa, which allows for open-hole compression and environmental knockdown [14].

Property	Al 7075-T6 (MAT 1)	Quasi-isotropic CFRP (MAT 2)
Young's modulus E	71,700 MPa	50,000 MPa
Poisson's ratio ν	0.33	0.28
Density ρ	2,810 kg/m^3	1,600 kg/m^3
Reference strength	F_{ty} = 480 MPa (design); 503 MPa typical	F_{tu} = 300 MPa (design)
Design allowable (Eq. 3.13)	320 MPa	200 MPa
Components	Spars, ribs, stringers, caps, flanges	Aerodynamic skins

Table 4.4 Material properties

4.6 Coordinate System, Units and Software

The model uses the aircraft Cartesian frame: X runs chordwise aft from the root leading edge, Y runs spanwise outboard from the root rib (y = 0) to the tip rib (y = 3,978 mm), and Z points upward. The solver uses the consistent unit system N–mm–tonne–MPa, so gravity is entered as g = 9,810 mm/s^2 and densities in tonne/mm^3.

Finite element work was carried out in ANSYS Mechanical APDL 2026 R1 in batch mode under the Student licence. Load pre-processing used MATLAB, and post-processing and file handling used Python 3.12 and Windows PowerShell.

4.7 Structural Configurations

Three layouts with different amounts of internal structure were defined so that the parametric script could be tested on more than one topology:

1) **Configuration A** reproduces the reference layout of Palumbo et al.: a three-spar torque box with 14 ribs and 8 stringers.
2) **Configuration B** removes the mid spar to give a conventional two-spar box, with 9 ribs and 8 stringers.
3) **Configuration C** reduces the framework further to 7 ribs, 2 spars and 6 stringers.

Parameter	Config A	Config B	Config C
Ribs / spars / stringers	14 / 3 / 8	9 / 2 / 8	7 / 2 / 6
Rib bays and pitch Δy	13 @ 306.00 mm	8 @ 497.25 mm	6 @ 663.00 mm
Front spar (x/C)	0.300	0.300	0.300
Mid spar (x/C)	0.764	Removed	Removed
Rear spar (x/C)	0.827	0.764	0.764
Stringer stations (x/C), each surface	0.409, 0.512, 0.607, 0.691	0.409, 0.512, 0.607, 0.691	0.409, 0.560, 0.691
Lightening holes per rib	7	6	5
Torque box cells	2	1	1

Table 4.5 Topology of the three structural configurations

All three configurations use the same section sizes (Table 5.2), so differences between them come from topology alone.

4.8 Verification Acceptance Criteria

Following the verification approach of Palumbo et al. [1] and the theory of Sections 3.6 and 3.7, three acceptance gates were set before any flight load was applied (Table 4.6).

Gate	Check	Acceptance criterion
1	Rigid-body modes (free-free)	First six modes are rigid-body modes with $f_{1\ldots6} < 1.0 \times 10^{-3}$ Hz
2	Mass conservation	Volume-integral mass equals solver mass matrix total ($\Delta = 0.0000\%$)
3	Gravity equilibrium	1g along X, Y and Z: reaction equals weight within 0.001%; strength ratio > 1

Table 4.6 Model verification acceptance gates

4.9 Refinements to Preliminary Assumptions

Several values from the preliminary data sheet were refined as the model and loads were developed (Table 4.7), so that the numbers in this report can be traced back to the Step 1 document.

Item	Preliminary value	Value used in this report
Al 7075-T6 strength basis	503 MPa (typical)	480 MPa design yield; σ_{allow} = 320 MPa
Positive limit load factor	+2.5	+2.5705 from Equation 3.1
Zero-lift pitching moment	$C_{m0} = 0$	0 for the isolated NACA 0012 tail; −0.06 for the wing-body in trim
Design dive speed	Not defined	$V_D = 1.25\ V_C$ = 312.5 kt EAS
Rib pitch	≈ 300 mm and 525 mm	306.00, 497.25 and 663.00 mm (exact)
Lightening holes per rib	6, 6, 4	7, 6, 5 (1 LE hole + box holes)
Loaded skin nodes	“7,000+” grid points	1,053 (A), 648 (B), 486 (C)

Table 4.7 Refinements between the preliminary assumptions and the final model

PROCEDURES

5.1 Step 1: Data Collection and Parameter Definition

Public data for the ATR 42-600 [3] and the requirements of CS-25 [2] were collected and combined with the tailplane geometry of Palumbo et al. [1] and handbook material data [5], [6]. The resulting parameters are those of Tables 4.2 to 4.4. They were entered as named parameters at the top of the APDL macro and of the MATLAB scripts, so that every later step reads them from a single source.

5.2 Step 2: Parametric APDL Script Development

5.2.1 Script Architecture

The master macro HTP_PARAMETRIC.mac builds the complete tailplane from parameters alone. It is divided into ten sequential blocks (Table 5.1) so that inputs, Boolean operations, meshing, boundary conditions and post-processing are kept separate, and no geometry repair or manual remeshing is needed when a parameter changes.

Block	Function	Main APDL commands
1	Inputs, configuration selector, sizing variables, allowables	*IF, *ELSEIF, *DIM
2	Rib stations, chord law, dihedral, NACA 0012 profile	*DO, TAN, SQRT
3	Spar-aware bays for lightening holes	*DO, *IF
4	Keypoints and spline boundary lines	K, L, BSPLIN
5	Areas: ribs, cutouts, skins, spar webs	AL, CYL4, WPOFFS, ASBA, CM
6	Topological clean-up and line extraction	NUMMRG, NUMCMP, LSLA
7	Element types, materials and beam sections	ET, MP, SECTYPE, SECOFFSET
8	Meshing, beam orientation, node merging	ESIZE, AMESH, LMESH, NUMMRG
9	Root clamp, skin node export, load input	D, *VWRITE, /INPUT
10	Solution and post-processing	SOLVE, PLNSOL, *GET, NSORT

Table 5.1 Modular structure of HTP_PARAMETRIC.mac

5.2.2 Geometry Generation

Block 2 computes the rib stations and the chord at each station. The chord varies linearly from root to tip and the N_{rib} ribs are spaced uniformly:

$$C(y) = 3{,}090 - 1{,}230 \cdot (y / 3{,}978) \ \text{[mm]} \tag{5.1}$$

$$y_i = (i - 1) \cdot s / (N_{rib} - 1), \quad i = 1, 2, \ldots, N_{rib} \tag{5.2}$$

For the 14-rib baseline this gives a pitch of 3,978 / 13 = 306.00 mm. Each station is raised by $y \cdot \tan 7.0°$ for dihedral, and the leading edge is moved aft by $X_{offset} = 1{,}230 \cdot (y / 3{,}978)$ mm so that the trailing edge stays straight. The airfoil follows Equation 3.8, with 25 points on

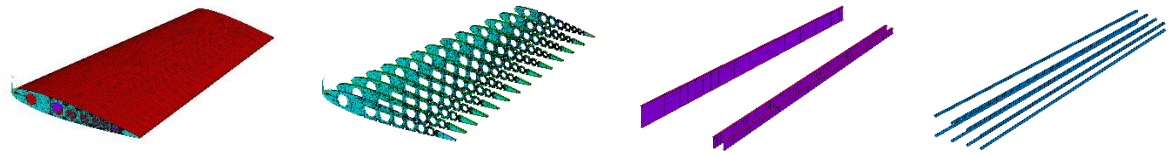
the upper surface mirrored to the lower surface (49 keypoints per rib). Block 4 joins the keypoints with splines, and Block 5 creates the rib, skin and spar web areas.

5.2.3 Elements and Sections

Plates are meshed with SHELL181 using full integration (KEYOPT(3) = 2), and stiffeners with BEAM188 using cubic shape functions (KEYOPT(3) = 3). Eight sections are defined (Table 5.2 and Figure 5.1) beam offsets place each section flush against the skin rather than through it.

Section	Component	Element	Dimensions [mm]	Material	Offset
SEC 1	Rib webs	SHELL181	t = 2.0	Al 7075-T6	Mid-plane
SEC 2	Spar webs	SHELL181	t = 3.0	Al 7075-T6	Mid-plane
SEC 3	Aerodynamic skins	SHELL181	t = 2.0	CFRP (QI)	Mid-plane
SEC 4	Hat stringers	BEAM188	30 × 20 × 15, t = 1.6	Al 7075-T6	Centroid CG_Y
SEC 5	Spar caps	BEAM188	45.0 × 2.5	Al 7075-T6	Z = −1.25
SEC 6	Interior rib flanges	BEAM188	40.0 × 1.0	Al 7075-T6	Z = −1.5
SEC 7	Root rib flange	BEAM188	20.0 × 1.0	Al 7075-T6	Y = −10, Z = −1.5
SEC 8	Tip rib flange	BEAM188	20.0 × 1.0	Al 7075-T6	Y = +10, Z = −1.5

Table 5.2 Structural sections and cross-section sizes



(a) (b) (c) (d)

Figure 5.1 Model Configuration A: (a) Part Model, (b) Ribs, (c) Spars, (d) Stringers.

5.2.4 Structural Detailing

Five detailing features were coded to match airframe practice [10]:

1) **Filleted mouseholes.** Stringers pass through the ribs in cutouts drawn with tangent fillets of radius 10 mm, which lowers the stress concentration factor from above 2.4 for a square cutout to about 1.4. The 56 mm cutout leaves 3 mm clearance on each side of the 50 mm stringer base.
2) **Safe pillar capping.** Lightening hole radius is 50% of the local rib height, limited so that at least 14 mm of web remains next to every mousehole: $R_{hole} \le (X_{mouse} - X_{hole}) \cdot C_i - HW - 14$ mm. Each rib has one leading-edge hole at x/C = 0.1535 and the rest inside the wingbox.
3) **Beam offsets.** The stringer offset equals the section centroid read with *GET, and the rib flange offset is $-(t_{flange} + t_{skin})/2 = -1.5$ mm, giving zero gap and zero penetration.
4) **Dual-zone flange orientation.** Rib flanges must tilt with the 7° dihedral, otherwise a 40 mm flange pierces the skin by 20 × sin 7° ≈ 2.44 mm; at the sharp nose the same tilt makes the flange curl forward. The tilt is therefore blended in linearly between x/C = 0.025 and 0.065.

5) **Single-sided boundary flanges.** The root and tip ribs use 20 mm flanges on the inboard side only (SEC 7 and SEC 8), so nothing projects outside the structure.



Figure 5.2 Elevation of Rib 2 (y = 306 mm) showing capped lightening holes and filleted mouseholes

5.2.5 Meshing and Connectivity

Skins and spar webs use a global element size of 100 mm, and rib webs use 15 mm to resolve the cutouts. Shells and beams are joined by shared lines and a node merge with a tolerance of 0.01 mm, so no constraint equations or contact elements are needed. With a smallest element edge of about 15 mm, the ratio 15.0 / 0.01 = 1,500 ensures that neighbouring nodes are never merged by accident. After every build, the companion macro VERIFY_VISUAL.mac captures 18 standard views so that geometric errors are found immediately.

5.2.6 Problems Solved During Development

Problem	Cause	Solution
Area ID recycling mesh crash	ASBA freed low area IDs that were reused by skins	Component-based selection (CM / CMSEL) instead of MAXD
Lightening holes misplaced	Working plane shifted along Z instead of Y	WPOFFS, 0, Y_STA(I), 0
Sliver elements at the nose	SMRTSIZE with spline splitting	Uniform ESIZE = 15 mm on rib areas
Singular stiffness matrix	Orphan nodes left after Booleans	Automatic node clean-up before solution
Missing nose flanges (Ribs 1–9)	Nose sagitta 16.19 mm exceeded 12 mm filter	Dual tolerance: 22 mm at nose, 12 mm elsewhere
Flange “beak” at the nose	3D cross-product combined slope and sweep	In-plane orientation for x/C < 0.025
Flange piercing the skin	No dihedral tilt across the wingbox	Dual-zone orientation blending
Truncated APDL expressions	Spaces around * read as a comment	Removed spaces around operators

Table 5.3 Main development problems and their solutions

5.3 Step 3: SFBM Load Pre-processing

5.3.1 Pipeline

Flight loads are usually given as spanwise shear force and bending moment (SFBM) distributions, whereas the FE solver needs nodal forces. A seven-part MATLAB–APDL pipeline was built to derive the loads and map them onto the model (Table 5.4).

Part	Routine	Method	Output
1	vn_diagram.m	CS-25 manoeuvre and gust boundaries (Section 3.1)	12 boundary flight points
2	tail_load.m	Pitch equilibrium at forward and aft CG (Section 3.2)	24 tail load cases
3	htp_sfbm.m	Schrenk's approximation (Section 3.3)	w(y), V(y), M(y)
4	HTP_PARAMETRIC.mac	Skin-only node export (MAT 2)	Node IDs and coordinates
5	htp_nodal_forces.m	Cubic load polynomial, tributary strips, lsqminnorm	APDL load files (.inp)
6	HTP_PARAMETRIC.mac	Root clamp and /INPUT of loads	Loaded static model
7	/POST1	Reaction extraction (PRRSOL)	Equilibrium audit

Table 5.4 Parts of the load pre-processing pipeline

5.3.2 Flight Envelope

With MTOW = 18,600 kg (41,006 lb), Equations 3.1 and 3.2 give:

$$n_{max} = 2.1 + 24{,}000 / (41{,}006 + 10{,}000) = 2.5705 \quad (5.3)$$

$$V_{S1} = 60.4 \text{ m/s} \approx 117.3 \text{ kt EAS}, \qquad V_A = 117.3 \times \sqrt{2.5705} = 188.1 \text{ kt EAS} \quad (5.4)$$

$$V_D = 1.25\, V_C = 1.25 \times 250 = 312.5 \text{ kt EAS} \quad (5.5)$$

The design gust speed is V_B = 182.96 kt EAS. Gust load factors were computed for 66, 50 and 25 ft/s gusts at V_B, V_C and V_D. The boundary points are listed in Table 5.5 and plotted in Figure 5.3.

Point	V [kt EAS]	n	Flight condition
A	188.10	+2.5705	Manoeuvre speed V_A at n_{max}
C	250.00	+2.5705	Cruise speed V_C at n_{max}
D	312.50	+2.5705	Dive speed V_D at n_{max}
E	312.50	0.0000	Dive speed at zero lift
F	250.00	−1.0000	Cruise speed at n_{min}
G	143.69	−1.0000	Negative stall meets n_{min}
Gust B±	182.96	+2.3833 / −0.3833	±66 ft/s gust at V_B
Gust C±	250.00	+2.4319 / −0.4319	±50 ft/s gust at V_C
Gust D±	312.50	+1.8950 / +0.1050	±25 ft/s gust at V_D

Table 5.5 Boundary flight points of the V-n diagram

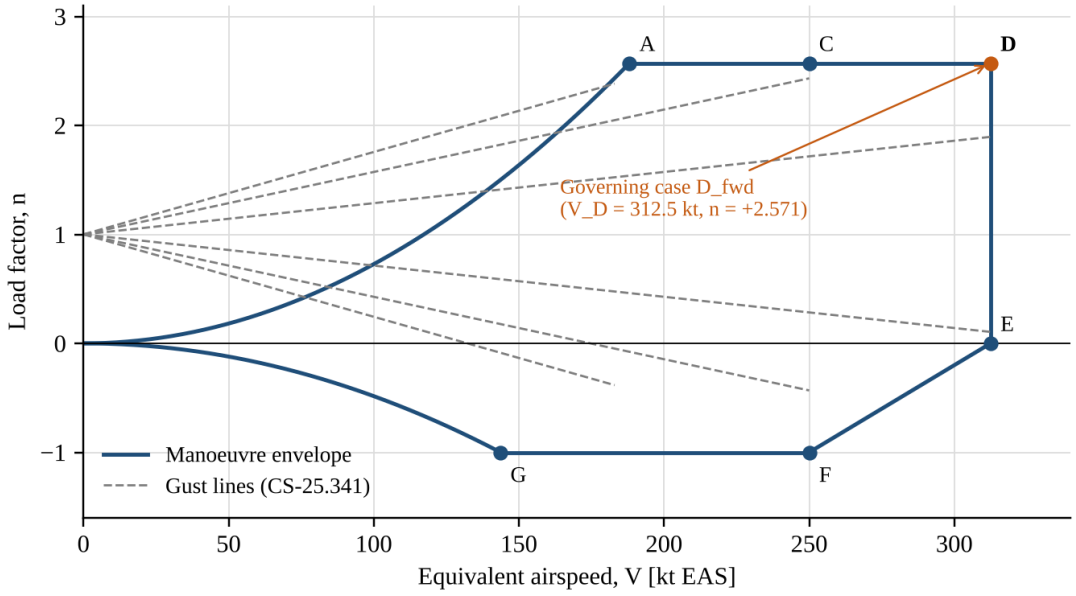


Figure 5.3 Manoeuvre and gust envelope of the ATR 42-600 (CS-25)

5.3.3 Balancing Tail Load

Equation 3.5 was evaluated for all flight points at the forward (16.8% $\bar{c}$) and aft (32.5% $\bar{c}$) CG limits, with the wing aerodynamic centre at 25% $\bar{c}$. The critical cases are listed in Table 5.6. Because dynamic pressure peaks at V_D (q = 15,829.3 Pa) and the forward CG gives the largest nose-down moment, case D_{fwd} governs.

Case	Point / CG	V [kt]	n	L_H [N]	P = L_H/2 [N]	Character
D_{fwd}	**D / forward**	**312.50**	**+2.5705**	**−18,285.7**	**−9,142.84**	**Governing download**
C_{fwd}	C / forward	250.00	+2.5705	−14,508.7	−7,254.35	Cruise manoeuvre
A_{fwd}	A / forward	188.10	+2.5705	−11,595.2	−5,797.60	Manoeuvre corner
A_{aft}	A / aft	188.10	+2.5705	+3,223.6	+1,611.80	Maximum upload

Table 5.6 Critical balancing tail load cases

As a hand check of case D_{fwd}: nW = 468,876 N, $(x_{cg} - x_{ac})\,\bar{c}$ = −0.187 m and $C_{m0}\,q\,S_w\,\bar{c}$ = −118,276 N·m, so L_H = −206,129 / 11.273 = −18,286 N, in agreement with the MATLAB result.

5.3.4 Spanwise Shear Force and Bending Moment

The half-tail load P = −9,142.84 N was distributed along the semi-span with Equations 3.6 and 3.7. The load centroid lies at y = 1.756 m (44.1% of the semi-span), giving V_{root} = −9,142.84 N and M_{root} = −16,058.0 N·m. Values at the Configuration A rib stations are listed in Table 5.7 and plotted in Figure 5.4.

Rib	y [m]	w(y) [N/m]	V(y) [N]	M(y) [N·m]
1	0.000	−2,851.3	−9,142.8	−16,058.0
2	0.306	−2,794.6	−8,279.1	−13,391.0
3	0.612	−2,738.7	−7,432.7	−10,985.7
4	0.918	−2,674.0	−6,605.0	−8,836.9
5	1.224	−2,600.0	−5,798.1	−6,940.7
6	1.530	−2,516.2	−5,015.0	−5,286.6
7	1.836	−2,421.5	−4,259.3	−3,867.7
8	2.142	−2,314.7	−3,534.6	−2,674.8
9	2.448	−2,193.8	−2,844.7	−1,698.8
10	2.754	−2,055.3	−2,194.2	−929.9
11	3.060	−1,893.7	−1,590.2	−351.4
12	3.366	−1,698.4	−1,041.0	−50.1
13	3.672	−1,442.7	−560.1	−4.4
14	3.978	−847.2	0.0	0.0

Table 5.7 Spanwise running load, shear force and bending moment for case D_{fwd}

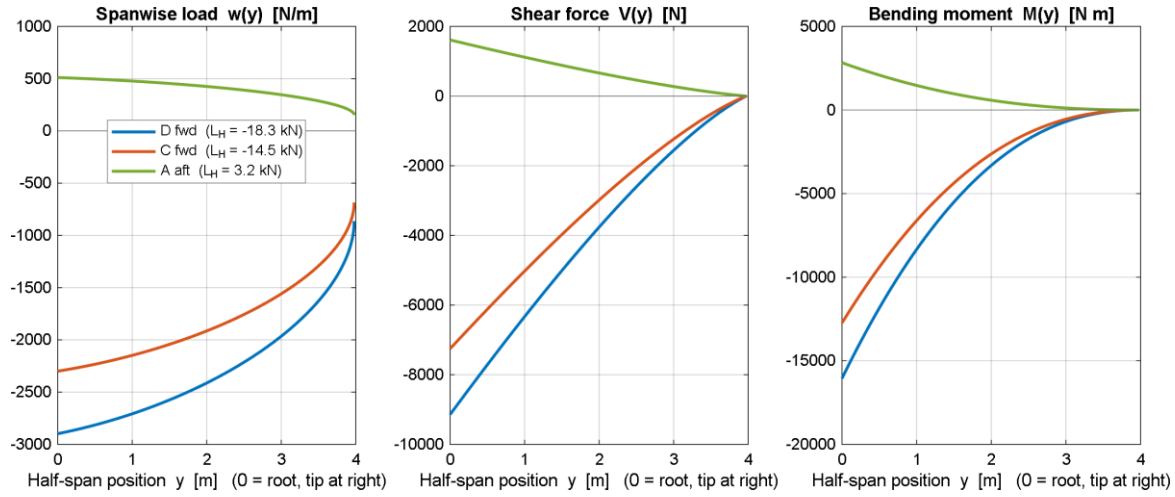


Figure 5.4 (a) Spanwise load, (b) shear force and (c) bending moment for case D_{fwd}

5.3.5 Skin Node Extraction

Block 9 exports node IDs and coordinates after the final node merge and renumbering, so that they match the solved database. Nodes are selected by material (ESEL, S, MAT,, 2 then NSLE), which picks only the corner nodes of skin shells, and the root station is skipped because forces there pass straight into the supports. The export gives 1,053 nodes for Configuration A, 648 for B and 486 for C.

5.3.6 Nodal Force Mapping

The running load is represented by a cubic polynomial in y/s. Each rib station g receives the load integrated over a strip reaching to the mid-bays on either side, so the whole semi-span is covered without gaps, and the station load is shared equally among its skin nodes:

$$F_{station}(g) = \int w(y)\,dy \text{ over } [y_{g-½}, y_{g+½}], \qquad F_{node} = F_{station}(g) / N_{nodes}(g) \tag{5.6}$$

The polynomial coefficients are fitted to the target shear and moment between stations with MATLAB lsqminnorm, with the root values V(0) = P and M(0) = M_{root} weighted by a

factor of 100. The forces are written as APDL commands (F, node, FZ, value) to htp_loads_D_fwd_cfg1–3.inp.

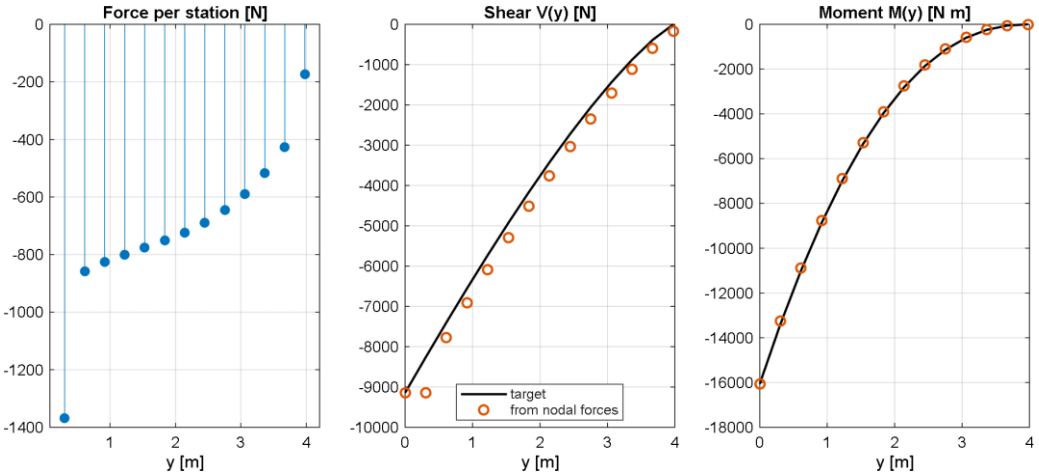


Figure 5.5 Target SFBM against the distribution reconstructed from nodal forces, Configuration A

5.3.7 Load Application and Equilibrium Audit

All degrees of freedom are fixed at the root rib, the configuration's load file is read with /INPUT and a linear static solution is run. The root reactions are then summed in /POST1. Table 5.8 shows that the applied load reaches the root without loss in every configuration.

Configuration	Loaded nodes	Applied ΣF_z [N]	Root reaction ΣR_z [N]	Error
A (14-3-8)	1,053	−9,142.84	+9,142.84	< 0.001%
B (9-2-8)	648	−9,142.84	+9,142.84	< 0.001%
C (7-2-6)	486	−9,142.84	+9,142.82	< 0.001%

Table 5.8 Applied load against root reaction

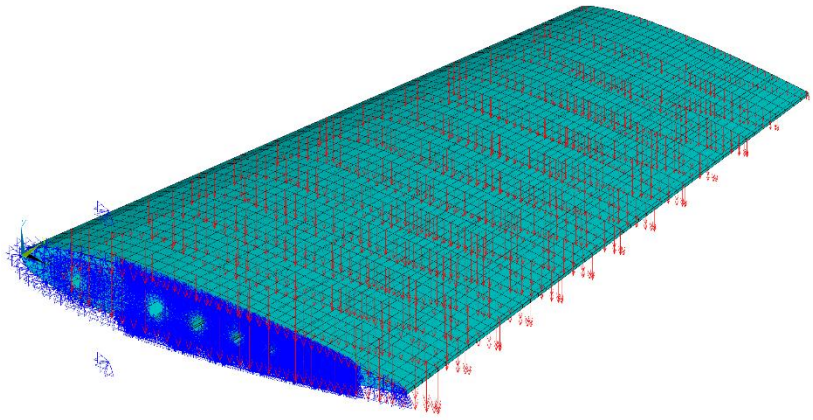


Figure 5.6 Configuration A applied nodal forces (F_z) and clamped root boundary condition

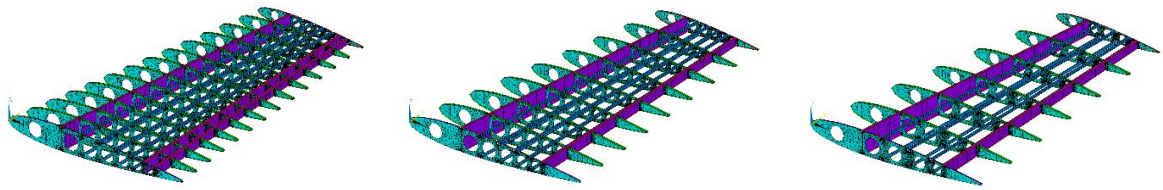
Because the load files refer to explicit node IDs, they stay valid when only thicknesses or section sizes change. A change of topology, mesh size or overall geometry renumbers the nodes, so Parts 4 and 5 must then be repeated.

5.4 Step 4: Configuration Setup

The integer CONFIG in Block 1 sets the number of ribs, spars, stringers and lightening holes (N_{rib}, N_{spar}, N_{str}, N_{cut}) for the layouts of Table 4.5. The change propagates automatically to the rib loops, spar stations, stringer lines, cutouts and component groups of every later block. For each configuration the skin nodes were exported again and a separate load file generated, so all three models carry the same total load. The resulting meshes are summarized in Table 5.9.

Quantity	Config A	Config B	Config C
Nodes	40,607	25,063	19,199
Elements (total)	63,460	38,251	29,815
SHELL181 / BEAM188	59,948 / 3,512	36,047 / 2,204	28,241 / 1,574
Use of 128,000-node licence limit	31.7%	19.6%	15.0%
Loaded skin nodes	1,053	648	486
Half-span mass	140.46 kg	121.89 kg	113.64 kg

Table 5.9 Finite element mesh census and half-span mass



(a) (b) (c)

Figure 5.7 Internal skeletons of (a) Configuration A, (b) Configuration B, (c) Configuration C

5.5 Step 5: Model Verification

The three models were run through the three gates of Table 4.6 with the verification macros in modal_analysis/. Following Palumbo et al. [1], the first six free-free modes were extracted with the Block Lanczos method and compared with the reference values.

5.5.1 Free-Free Modal Analysis

With all constraints removed, each model was solved for its first six modes. Table 5.10 lists the frequencies of the three configurations against the reference values of Palumbo et al. [1], together with the mode shapes of Configuration A (Figure 5.8), plotted as signed displacement components in the same way as the reference paper.

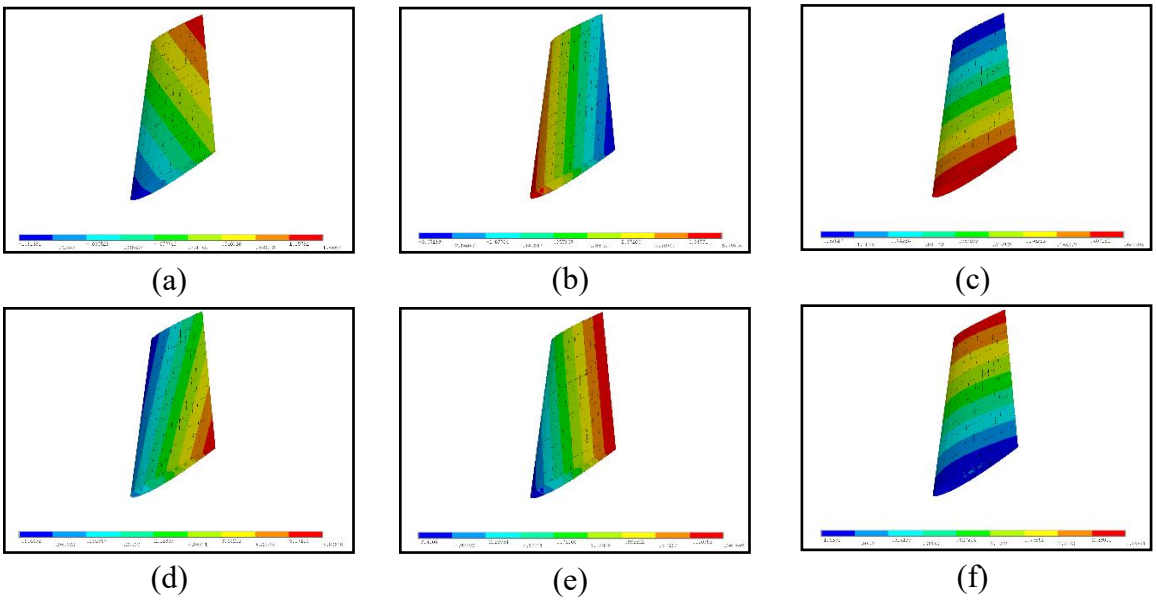


Figure 5.8 (a) Mode 1, (b) Mode 2, (c) Mode 3, (d) Mode 4, (e) Mode 5, (f) Mode 6.

Mode	Palumbo et al. [1] [Hz]	Config A [Hz]	Config B [Hz]	Config C [Hz]
1	0	0	0	0
2	0	0	0	0
3	0	1.82×10^{-5}	3.87×10^{-5}	0
4	2.46×10^{-5}	9.51×10^{-5}	6.97×10^{-5}	0
5	3.79×10^{-5}	1.30×10^{-4}	1.36×10^{-4}	9.78×10^{-5}
6	4.92×10^{-4}	2.46×10^{-4}	1.97×10^{-4}	2.41×10^{-4}

Table 5.10 Free-free rigid-body frequencies and mode shapes (Configuration A)

All six modes of every configuration have frequencies below 2.47×10^{-4} Hz, four times lower than the gate limit and of the same order as the reference values, which confirms that every skin panel, web, stringer and flange is connected. The mode shapes show the straight, parallel contour bands predicted by Equation 3.11 and seen in the reference paper, as expected for rigid-body motion.

5.5.2 Mass Audit

Mass was computed from element volumes (ETABLE, VOLU) using Equation 3.12 and, independently, from the solver mass matrix (ELEM, MTOT). The two results agree exactly (Table 5.11); the breakdown by component is shown in Figure 5.9.

Component	Config A [kg]	Config B [kg]	Config C [kg]
Composite skins (SEC 3)	63.665	63.665	63.665
Rib webs (SEC 1)	28.835	18.928	15.337
Spar webs (SEC 2)	19.064	15.209	15.209
Hat stringers (SEC 4)	15.083	15.083	11.314
Spar caps (SEC 5)	7.671	5.133	5.133
Rib flanges (SEC 6–8)	6.147	3.874	2.981
Total, volume method	**140.463**	**121.890**	**113.638**
Total, solver mass matrix	140.463	121.890	113.638
Discrepancy	0.0000%	0.0000%	0.0000%

Table 5.11 Component mass breakdown and dual-method audit (half-span)

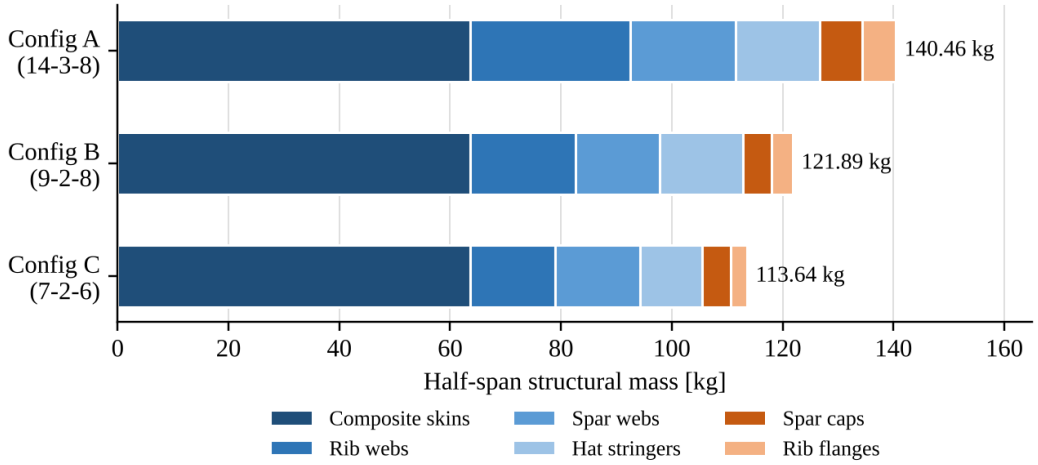


Figure 5.9 Half-span mass breakdown by component

5.5.3 Gravity Load Verification

Static analyses under 1g were run separately in the X, Y and Z directions with the root rib clamped, following Tables 4–6 of the reference paper. The root reaction in the loaded direction matches the model weight in every case (Table 5.12). Under 1g in Z the maximum displacement is about 1 mm and the largest stress is below 2 MPa, so the strength ratio (allowable stress divided by applied stress) is far above one.

Quantity	Config A	Config B	Config C
Model mass [ton]	0.14046	0.12189	0.11364
Mass × 1g [N]	−1,377.946	−1,195.743	−1,114.784
Reaction, 1g in X [N]	−1,377.946	−1,195.743	−1,114.784
Error, X [%]	0.6×10^{-7}	0.4×10^{-6}	0.6×10^{-6}
Reaction, 1g in Y [N]	−1,377.946	−1,195.743	−1,114.784
Error, Y [%]	0.5×10^{-6}	0.6×10^{-6}	0.5×10^{-6}
Reaction, 1g in Z [N]	−1,377.946	−1,195.743	−1,114.784
Error, Z [%]	0.9×10^{-6}	0.7×10^{-6}	0.1×10^{-6}
1g Z: max displacement [mm]	1.080	0.971	0.938
1g Z: max stress [MPa]	1.98	1.75	1.61
Strength ratio (480 MPa / σ_{max})	242.3	274.3	297.4

Table 5.12 Gravity load verification

5.5.4 Verification Summary

Gate	Criterion	Config A	Config B	Config C
1 Rigid-body modes	max $f_{1\ldots6} \leq 10^{-3}$ Hz	2.46×10^{-4}	1.97×10^{-4}	2.41×10^{-4}
2 Mass conservation	$\Delta = 0.0000\%$	0.0000%	0.0000%	0.0000%
3 Gravity equilibrium	error < 0.001%	$< 10^{-6}\%$	$< 10^{-6}\%$	$< 10^{-6}\%$
Overall		**PASS**	**PASS**	**PASS**

Table 5.13 Verification gate summary

CONCLUSION AND DISCUSSION

6.1 Conclusion

This report has presented the development, loading and verification of a parametric finite element model of a horizontal tailplane, with flight loads reconstructed for the ATR 42-600 under EASA CS-25. Against the three objectives:

1) **Parametric model.** A single APDL macro generates three complete internal layouts, including filleted mouseholes, capped lightening holes and flush beam offsets, without any manual remodelling. The largest model (Configuration A, 40,607 nodes) uses 31.7% of the Student licence limit.
2) **Load reconstruction and mapping.** The governing limit case is the dive-speed manoeuvre at the forward CG limit, giving a half-tail download of 9,142.84 N and a root bending moment of 16,058 N·m. Mapped onto 1,053, 648 and 486 skin nodes, the load is returned at the root with an error below 0.001% in every configuration.
3) **Model verification.** All three models pass all three gates: six rigid-body modes below 2.5×10^{-4} Hz, exact agreement between the two mass calculations, and gravity reactions within 10^{-6}% of the model weight with strength ratios above 240.

The models are therefore correctly connected, carry the correct mass and satisfy equilibrium under both gravity and flight loads, and can be used with confidence as the basis for stress and design studies.

6.2 Discussion

Load reconstruction. The hand check of Equation 3.5 reproduces the MATLAB tail load to within 1 N, which confirms that the pitch-equilibrium implementation is correct. The forward CG limit governs because both the CG offset and the wing-body pitching moment push the tail load downward, and the effect grows with dynamic pressure.

Load mapping. Two errors found during development show why the equilibrium audit is essential. Selecting nodes by coordinates alone picked up internal nodes of the cubic BEAM188 elements, which are not connected to the shells, and 44% of the load disappeared (a root reaction of 5,102 N against 9,143 N applied). Applying station loads as point forces without tributary strips shifted the resultant outboard and overestimated the root moment by 17%. Both errors would have been invisible without the reaction check.

Comparison with the reference. The rigid-body frequencies are of the same order as those of Palumbo et al., and the Configuration A mass of 140.46 kg is 15.4% lower than their 166.02 kg. The difference comes from the lighter Hat stringer (30 × 20 × 15 mm, t = 1.6 mm) and compact spar caps (45 × 2.5 mm) used here rather than from a modelling error, since the mass audit agrees exactly.

6.3 Limitations

1) Only load case D_{fwd} was reconstructed and applied. The upload case A_{aft} and the gust cases were identified but not applied.
2) Forces act purely in the Z direction; local torsion from the offset between the aerodynamic centre and the elastic axis is not included.
3) The composite skin is an equivalent isotropic shell rather than a layered laminate.
4) No mesh convergence study was performed; the mesh follows literature practice.
5) The verification confirms that the models are correct, but stresses, deflections and stability under flight load have not yet been assessed.

6.4 Recommendations for Future Work

1) Run linear static and buckling analyses of the three configurations under case D_{fwd}, followed by the upload case A_{aft}, using the verified models and load files.
2) Extend the load set to the full CS-25 envelope, including gusts and unsymmetrical cases.
3) Model the skin as a layered SHELL181 laminate with ply-level failure criteria.
4) Perform a mesh convergence study on stresses and frequencies.
5) Use the node-invariant load files, which remain valid for any change of thickness, as the basis for a later sizing study.

REFERENCES

[1] C. Palumbo et al., “A parametric APDL-based workflow for efficient structural design of a civil aircraft horizontal tailplane,” *Engineering Proceedings*, vol. 133, no. 15, 2026.

[2] European Union Aviation Safety Agency, *Certification Specifications and Acceptable Means of Compliance for Large Aeroplanes (CS-25)*, Amendment 27, 2023.

[3] ATR Aircraft, *ATR 42-600 Aircraft Characteristics: Airport and Maintenance Planning*, Blagnac, France, 2020.

[4] E. N. Jacobs, K. E. Ward and R. M. Pinkerton, “The characteristics of 78 related airfoil sections from tests in the variable-density wind tunnel,” NACA Report No. 460, 1933.

[5] ASM International Handbook Committee, *ASM Handbook, Vol. 2: Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*. Materials Park, OH: ASM International, 1990.

[6] C. Lundsgaard-Larsen et al., quasi-isotropic carbon/epoxy laminate properties for civil aircraft structures, 2012.

[7] O. Schrenk, “A simple approximation method for obtaining the spanwise lift distribution,” NACA Technical Memorandum No. 948, 1940.

[8] D. P. Raymer, *Aircraft Design: A Conceptual Approach*, 6th ed. Reston, VA: AIAA, 2018.

[9] T. H. G. Megson, *Aircraft Structures for Engineering Students*, 6th ed. Oxford: Butterworth-Heinemann, 2017.

[10] M. C. Y. Niu, *Airframe Structural Design*, 2nd ed. Hong Kong: Conmilit Press, 1999.

[11] K. J. Bathe, *Finite Element Procedures*. Englewood Cliffs, NJ: Prentice Hall, 2006.

[12] R. W. Clough and J. Penzien, *Dynamics of Structures*, 3rd ed. Berkeley, CA: Computers & Structures, 2003.

[13] Federal Aviation Administration, *Metallic Materials Properties Development and Standardization (MMPDS-01)*, 2003.

[14] SAE International, *Composite Materials Handbook CMH-17, Vol. 3: Polymer Matrix Composites: Materials Usage, Design, and Analysis*, Rev. G, 2012.